

Victoria Puck-Karam

Software Engineer II — Backend & AI Systems
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SUMMARY

Software Engineer II building scalable backend systems, distributed microservices, and AI-powered platforms in AWS. Experience delivering production RAG systems, event-driven architectures, and developer platforms at JPMorgan Chase & Co.

EXPERIENCE

JPMorgan Chase & Co.

Wilmington, DE

Software Engineer II

January 2026 – Present

- Recognized in the top 5% of global early-career software engineers and fast-tracked for promotion based on performance and impact.
- Architected an internal LLM platform using RAG architecture and MCP servers, enabling scalable semantic retrieval across enterprise knowledge system to accelerate incident root-cause analysis.
- Migrated a legacy monolith to a cloud-native microservices architecture, improving system reliability and deployment velocity.
- Scaled internal developer platform to support 922% user growth, increasing infrastructure capacity, and improving system reliability.

Software Engineer I

August 2024 – January 2026

- Delivered full-stack internal automation tools with Java Spring, Terraform, Python, and AWS, eliminating bottlenecks in the Home Lending loan delivery pipeline and saving hundreds of engineering hours annually.
- Translated complex regulatory and financial logic into backend automation using JS; deployed solutions that saved millions in compliance costs.
- Built and maintained resilient CI/CD infrastructure with Spinnaker and Jenkins, integrating contract, performance, and chaos testing to ensure safe, stable production releases.
- Implemented advanced observability tooling with CloudWatch, enabling real-time performance monitoring and reducing mean time to resolution (MTTR).
- Developed a privileged access management system for a SaaS platform, automating configurations and reducing deployment time across environments.

Software Engineering Consultant

September 2023 – November 2023

Out2Win

State College, PA

- Developed scalable Python AWS Lambda pipelines leveraging Playwright for automated web scraping, optimizing data ingestion for the startup's AI-powered athlete marketing system.
- Designed SQL data schemas to store and query structured athlete information, supporting machine learning workflows and analytics.
- Integrated social media scrapers for Instagram and TikTok, collecting engagement and follower data to enrich athlete profiles and enhance predictive marketing models.

Software Engineer Intern

June 2023 – August 2023

JPMorgan Chase & Co.- Data Technology

Wilmington, DE

- Engineered a high-throughput Java Spring Boot microservice to decrypt, parse, and store events from multiple data streams; integrated Kafka consumers to process over 10,000 messages per minute with real-time ingestion and persistence.
- Spearheaded the development of a Kubernetes-hosted Java back-end service for data pipeline management, alleviating a critical pain point for 20+ developer teams.
- Implemented JUnit unit tests and Cucumber integration, achieving a test coverage rate of 98.2%.
- Orchestrated deployment of a production-ready Java microservice through Jenkins and Kubernetes.

Data Science Intern

May 2022 – August 2022

MAXAR Technologies

Westminster, CO

- Surfacing actionable insights from large-scale historical sales and geospatial data sets and visualizing data using an ESRI dashboard to advise optimization and effective data-driven decision making essential to intelligence customers and in advising the sales execution teams.

- Executing the business understanding step of the data science life cycle by facilitating interdisciplinary communication to translate client, product manager, and sales team asks into executable technical tasks.
- Automating robust ETLs (Python & SQL) using an Apache Airflow server to ingest data from S3 buckets into PostgreSQL RDS.
- Deriving data understanding through performing statistical analysis using Scikit learn, PyMC and NumPy and producing preliminary visualizations with matplotlib, Seaborn, and PyPlot, as a precursor to modeling and finalized visualization.

PROJECTS

- CMCI Index** | *R, MatLab, MCMC, GARCH, Kalman Filtering* January 2023 – June 2024
- Utilized advanced methodologies: temporal and geographical expansion, Principal Component Analysis (PCA), Markov Chain Monte Carlo (MCMC) modeling with Stein Thinning, Kalman Filtering and GARCH Volatility Model
 - Optimized index offers a comprehensive understanding of the U.S. economy, particularly post-COVID-19
 - Contributed to literature on proxy indexes for consumer sentiment by refining factor model, assessing predictive power, and uncovering symmetries between consumer sentiment and macroeconomic factors
 - Developed a front-end dashboard to display the CMCI using Apache Airflow and Apache Superset
- Website Accessibility Project** | *Python, BeautifulSoup, Web Scraping* November 2020 – May 2022
- Developing Python algorithms to automate data analytics of elements in Digital Healthcare to draw insights about the accessibility of essential healthcare services.
 - Utilizing web-harvesting technology to extract data from 10,000 US hospitals' user interfaces
 - Normalizing and wrangling web-scraped data for analysis
- Spot-Suggest** | *Python, Scikit-Learn, PyTorch, TensorFlow* April 2021 – August 2021
- Deployed and trained a machine learning algorithm utilizing data from the Spotify API.
 - Utilized cosine similarity functionality to determine mathematical similarity of MIDI files
- X-READ** | *Python, TensorFlow, Keras* August 2019 – July 2020
- Developed CNN Machine Learning algorithm in Python3 using TensorFlow and Keras libraries within an Anaconda environment.
 - Automated the diagnosis of common Cardiothoracic illnesses from X-RAY scans from the NIH database.
 - Achieved an average accuracy of 65-75%, with the highest precision in diagnosing X-rays with visible masses reaching 91% accuracy

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, R, MATLAB

AI Platforms & LLM Systems: Generative AI, MCP Servers (Model Context Protocol), Retrieval-Augmented Generation (RAG), LLM Applications, Vector Search (OpenSearch)

Backend & Distributed Systems: Spring Boot, REST APIs, Microservices, Distributed Systems, Event-Driven Architecture

Cloud & Infrastructure: AWS (Lambda, ECS, S3, RDS, CloudWatch), Terraform (IaC), Docker, Kubernetes

Data Infrastructure: Kafka, Redis (Caching), DynamoDB (NoSQL), PostgreSQL, ETL Pipelines

DevOps, CI/CD & Reliability: CI/CD Pipelines (Jenkins, Spinnaker), Observability (Datadog, Dynatrace, Splunk), Testing (Cypress, Jest, Pact, Selenium, Gremlin, BlazeMeter)

EDUCATION

The Pennsylvania State University	University Park, PA
<i>Bachelor of Science in Computational Data Science</i>	<i>Aug. 2020 – May 2024</i>
College of Electrical Engineering and Computer Science (EECS)	
Minors in Mathematics and Engineering Leadership Development	